

University of Yaoundé I
Department of Physics, Faculty of Science
P.O. Box 812, Yaoundé, Cameroon
myke-vital.sao@facsciences-uy1.cm

August 2, 2026

The Editors
Journal of Cheminformatics
BioMed Central

Dear Editors,

We are pleased to submit our manuscript, **“Pareto-guided Monte Carlo tree search explores multi-objective trade-offs missed by scalar antimalarial generation”**, for consideration as a Research Article in the *Journal of Cheminformatics*.

The work addresses a recurring limitation of de novo molecular design: scalar reward aggregation that conceals trade-offs between potency, synthetic accessibility and drug-likeness. We introduce a Pareto-guided Monte Carlo tree search (MCTS) agent that maintains a non-dominated front across MPO, SYBA and SA scores, guided by a scaffold-aware fragment policy.

A four-method, 20-seed benchmark under a fixed oracle budget yields a deliberately honest ranking: random search achieves the highest mean scalar reward (0.734), followed closely by MCTS at its screened optimal configuration (0.728; mean gap 0.006), a genetic algorithm (0.703) and deterministic greedy search (0.615). MCTS is therefore competitive on single-objective performance, and is the only method that systematically returns a multi-objective Pareto front of non-dominated candidates (four solutions, hypervolume 1.237) spanning the potency–accessibility trade-off, including concave regions that scalar optimisation misses. Component ablations identify the chemistry-informed policy and the Pareto front itself as the dominant performance drivers.

We believe this transparent, methods-focused contribution aligns closely with the scope of the *Journal of Cheminformatics*: rigorous benchmarking, open deposition of code and molecule sets, and honest reporting of both strengths and limitations of a multi-objective search strategy. All data and code will be deposited under the MIT licence.

This work is original, is not under consideration elsewhere, and all authors have approved the submission.

We thank you for your consideration.

Myke Vital Sao Temgoua
Corresponding Author

Suggested reviewers:

- Dr. Jan H. Jensen (graph-based genetic algorithms / MCTS for chemical space)
- Prof. Gisbert Schneider (generative molecular design)
- Dr. Ola Engkvist (de novo design, multi-objective optimisation)